CBSE Class 12 Physics — Moving Charges and Magnetism

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (Objective/VSA)

Q1. An electron is moving along the positive x-axis in a uniform magnetic field which is parallel to the positive y-axis. In what direction will the magnetic force be acting on the electron? [PYQ 2023] Very High

Q2. An ammeter of resistance $0.81\ \Omega$ reads up to $1\ \mathrm{A}$. What is the value of the required shunt to safely increase its range to $10\ \mathrm{A}$? [PYQ 2023] High

Q3. An electron with orbital angular momentum L moving around the nucleus has a magnetic dipole moment. What is its magnitude? [PYQ 2023] Very High

Q4. A current-carrying straight wire kept in a uniform magnetic field will experience a maximum magnetic force when the angle between the wire and the magnetic field is? [PYQ 2021] High

Q5. A straight conducting rod of length l and mass m is suspended in a horizontal plane by a pair of flexible strings in a uniform magnetic field of magnitude B. To completely remove the tension in the supporting strings, what must be the magnitude of the current in the wire? [PYQ 2021] High

Q6. An electron is released from rest in a region of uniform electric and magnetic fields acting strictly parallel to each other. What path will the electron follow? [PYQ 2020] Very High

Q7. A charged particle, after being accelerated through a potential difference V, enters a uniform magnetic field and moves in a circle of radius r. If the accelerating voltage V is doubled, what will be the new radius of the circle? [PYQ 2020] High

Q8. The time period of a charged particle undergoing circular motion in a uniform magnetic field is strictly independent of which of its kinematic properties? [PYQ 2020] Very High

Q9. A long straight wire of radius a carries a steady current I. The current is uniformly distributed across its cross-section. What is the ratio of the magnitude of the magnetic field at a distance $a/2$ to that at a distance $2a$ from the axis? [PYQ 2023] High

Q10. Two wires of the exact same length are shaped into a square of side a and a circle of radius r. If they carry the same current, what is the ratio of their magnetic dipole moments? [PYQ 2021] Medium

Q11. The magnetic field at the centre of a current-carrying circular loop of radius R is B_1. The magnetic field at a point on its axis at a distance R from the centre is B_2. Find the ratio B_1/B_2. [PYQ 2021] High

Q12. Two concentric and coplanar circular loops P and Q have their radii in the ratio $2:3$. Loop Q carries a current of $9\ \mathrm{A}$ in the anticlockwise direction. For the net magnetic field to be exactly zero at their common centre, what current must loop P carry? [PYQ 2022] Very High

Q13. If a standard ammeter is to be used in place of a voltmeter to measure potential difference, how must we modify it? [PYQ 2021] High

Q14. The magnetic dipole moment of a current-carrying coil does NOT depend upon: (a) number of turns, (b) cross-sectional area, (c) current flowing, or (d) material of the turns. [PYQ 2020] Medium

Q15. A proton and an alpha particle enter a uniform magnetic field perpendicularly with the exact same speed. What is the ratio of their respective time periods of revolution ($T_p : T_\alpha$)? [Practice] High

Q16. Write the mathematical expression, in vector form, for the Lorentz magnetic force $\vec{F}$ acting on a charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$. [PYQ 2014] Very High

Q17. Define one Tesla using the expression for the magnetic force acting on a moving charged particle. [PYQ 2014] High

Q18. Assertion (A): When the radius of a circular wire carrying a steady current is doubled, its magnetic dipole moment becomes four times. **Reason (R):** The magnetic dipole moment of a current-carrying loop is directly proportional to the area of the loop. [PYQ 2021] Very High

Q19. Assertion (A): Two parallel wires carrying steady currents in the same direction attract each other. **Reason (R):** The magnetic field produced by one wire exerts a Lorentz force on the moving charges in the second wire, directed towards the first wire. [Practice] High

Q20. Assertion (A): The higher the range of an ammeter, the lower its overall electrical resistance. **Reason (R):** To increase the measuring range of an ammeter, an additional low-resistance shunt is connected in parallel to it. [PYQ 2021] Very High

2-Mark Questions

Q21. Two identical circular loops P and Q, each of radius $5\ \mathrm{cm}$, are lying in mutually perpendicular planes with a common centre. Find the magnitude and direction of the net magnetic field at their centre if they carry currents of $3\ \mathrm{A}$ and $4\ \mathrm{A}$ respectively. [PYQ 2023] Very High

Q22. A closely wound solenoid of 2000 turns and cross-sectional area $1.6 \times 10^{-4}\ \mathrm{m}^2$, carrying a current of $4.0\ \mathrm{A}$, is suspended through its centre allowing it to turn in a horizontal plane. What is the magnetic dipole moment associated with the solenoid? [PYQ 2022] High

Q23. A straight horizontal wire of mass $200\ \mathrm{g}$ and length $1.5\ \mathrm{m}$ carries a current of $2\ \mathrm{A}$. It is suspended motionless in mid-air by a uniform horizontal magnetic field B. What is the magnitude of the magnetic field? (Take $g = 9.8\ \mathrm{m/s^2}$). [PYQ 2015] High

Q24. Write the specific condition under which an electron will move entirely undeflected in the presence of crossed (mutually perpendicular) electric and magnetic fields. [PYQ 2014] Very High

Q25. A proton, a deuteron, and an alpha particle, after being accelerated through the same potential difference V, enter a uniform magnetic field perpendicular to their paths. Find the ratio of the radii of the circular paths described by the proton and the alpha particle. [PYQ 2019] Very High

Q26. Define the term 'current sensitivity' of a moving coil galvanometer. State its SI unit. [PYQ 2020] Very High

Q27. An ammeter of internal resistance $0.8\ \Omega$ can accurately measure a current up to $1.0\ \mathrm{A}$. Calculate the exact value of the shunt resistance required to convert this ammeter to measure a current up to $5.0\ \mathrm{A}$. [PYQ 2020] High

Q28. Two long straight parallel wires carry a current of $5\ \mathrm{A}$ each and are kept at a separation of $1\ \mathrm{m}$. If the currents are flowing in the same direction, calculate the force per unit length between them. Is it attractive or repulsive? [PYQ 2022] High

Q29. A square coil of side $10\ \mathrm{cm}$ consists of 20 turns and carries a current of $12\ \mathrm{A}$. The coil is suspended vertically and the normal to the plane of the coil makes an angle of 30° with the direction of a uniform horizontal magnetic field of magnitude $0.80\ \mathrm{T}$. Calculate the magnitude of torque experienced by the coil. [PYQ 2022] Medium

Q30. An electron with charge $-e$ and mass m travels at a constant speed v in a plane perpendicular to a uniform magnetic field B. The electron follows a circular path of radius R. In a time t, the electron travels halfway around the circle. What is the exact amount of work done by the magnetic field on the electron? [PYQ 2021] High

Q31. A proton and an electron travelling along parallel paths enter a region of uniform magnetic field acting perpendicular to their paths. Which of them will move in a circular path with a higher frequency? Give the mathematical reason. [PYQ 2018] High

Q32. Briefly explain why a high resistance is always connected in series with a galvanometer when converting it into a voltmeter. [PYQ 2014] Medium

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3-Mark Questions

Q33. State the Biot-Savart law and express it in vector form. Two identical circular coils, P and Q, each of radius R, carrying currents $1\ \mathrm{A}$ and $\sqrt{3}\ \mathrm{A}$ respectively, are placed concentrically and perpendicular to each other in the XY and YZ planes. Find the magnitude of the net magnetic field at their centre. [PYQ 2017] Very High

Q34. A given figure shows a long straight wire of a circular cross-section of radius a carrying a steady current I. The current is uniformly distributed across this cross-section. Using Ampere's circuital law, calculate the magnetic field in the regions: (i) $r < a$, and (ii) $r > a$. [PYQ 2023] Very High

Q35. A proton, a deuteron, and an alpha particle are accelerated through the same potential difference V and then subjected to a uniform magnetic field perpendicular to their directions of motion. Compare (i) their kinetic energies, and (ii) if the radius of the path described by the proton is $5\ \mathrm{cm}$, determine the radii of the paths described by the deuteron and the alpha particle. [PYQ 2019] High

Q36. A charged particle q moving with velocity v enters a uniform magnetic field B inclined at an angle of 30° to the direction of its motion. Trace the trajectory of the particle, explain why it follows this path, and deduce the mathematical expression for its pitch. [PYQ 2015] Very High

Q37. Two long straight parallel conductors carry steady currents I_1 and I_2 separated by a distance d. If the currents are flowing in the same direction, show mathematically how the magnetic field set up by one produces an attractive force on the other. Obtain the expression for this force per unit length. Hence, define one Ampere. [PYQ 2016] Very High

Q38. State Biot-Savart's law. Use this law to derive the expression for the magnetic field due to a current-carrying circular loop of radius R at a point P distant x from its centre along the axis of the loop. [PYQ 2016] High

Q39. Draw the magnetic field lines due to a steady current passing through a long solenoid. Use Ampere's circuital law to obtain the expression for the magnetic field deep inside a long solenoid having n turns per unit length. [PYQ 2014] High

Q40. Two long straight parallel wires A and B, separated by a distance d, carry equal current I flowing in the same direction. (i) Find the magnetic field at a point P situated exactly midway between them. (ii) Show graphically the variation of the magnetic field with distance x ($0 < x < d$) from wire A. [PYQ 2020] High

Q41. A square-shaped current-carrying loop MNOP is placed near a straight long current-carrying wire AB in the same plane. If the loop experiences a net force F directed towards the wire, explain qualitatively how this force arises by analyzing the magnetic forces on the individual arms of the square loop. [PYQ 2020] Medium

Q42. Give two explicit reasons to explain why a galvanometer cannot be used directly as an ammeter to measure the value of the current in a given circuit. Explain how it is practically converted into an ammeter. [PYQ 2019] Very High

5-Mark Questions (Long Answer/Case-Study)

Q43. (A) State the underlying principle of a moving coil galvanometer. (B) Derive the mathematical expression for the deflecting torque acting on the current-carrying rectangular coil of the galvanometer when placed in a magnetic field. (C) What is the specific purpose of employing a radial magnetic field in the moving coil galvanometer? How is it physically achieved? [PYQ 2020] Very High

Q44. (A) Derive the expression for the torque acting on a rectangular current-carrying loop placed in a uniform magnetic field $\vec{B}$. Indicate the direction of the torque. (B) In which orientations will the coil be in (i) stable equilibrium, and (ii) unstable equilibrium? Explain. (C) A square shaped plane coil of area $100\ \mathrm{cm}^2$ with 20 turns carries a steady current of $5\ \mathrm{A}$. It is placed in a uniform magnetic field of $0.2\ \mathrm{T}$. Calculate the maximum torque it can experience. [PYQ 2019] Very High

Q45. (A) Write the expression for the Lorentz force $\vec{F}$ acting on a particle of mass m and charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$. (B) Under what specific conditions will it move in (i) a purely circular path, and (ii) a helical path? (C) Show mathematically that the kinetic energy of a charged particle moving exclusively in a uniform magnetic field remains strictly constant. [PYQ 2017] High

Q46. (A) Two infinitely long straight wires A_1 and A_2 carrying currents I and $2I$ respectively flowing in the same direction are kept d distance apart. Where exactly should a third straight wire A_3 carrying a current of $1.5I$ be placed between A_1 and A_2 so that it experiences zero net magnetic force? (B) Does the net force acting on A_3 depend on the direction or magnitude of the current flowing through it? Justify. [PYQ 2019] Very High

Q47. (A) Write any two important points of similarities and differences each between Coulomb's law for the electrostatic field and Biot-Savart's law for the magnetic field. (B) Apply the Biot-Savart law to find the expression for the magnetic field at the exact centre of a circular loop of radius R carrying steady current I. [PYQ 2015] Medium

Q48. (A) Define Voltage Sensitivity and Current Sensitivity of a moving coil galvanometer. (B) Deduce the relationship between voltage sensitivity and current sensitivity. (C) A student increases the number of turns in the galvanometer coil to strictly double its current sensitivity. Does the voltage sensitivity also double? Explain why or why not. [PYQ 2019/Practice] High

Q49. An alpha particle and a proton are accelerated from rest through the exact same potential difference V. They then enter a region of uniform magnetic field B acting perpendicular to their velocity vectors. (A) Find the ratio of their final kinetic energies. (B) Find the ratio of their speeds as they enter the field. (C) Find the ratio of the radii of the circular paths they describe. (D) Find the ratio of their time periods of revolution. [Practice] High

Q50. Case Study: Galvanometer Conversions A moving coil galvanometer is a highly sensitive instrument used to detect small currents, but it has a fixed coil resistance R_g and gives full-scale deflection at a very small current I_g. (a) To convert this galvanometer into an ammeter of a larger range 0 to I, a shunt resistance S is used. Derive the formula for S in terms of I, I_g, and R_g. (b) To convert it into a voltmeter of range 0 to V, a series resistance R is used. Derive the formula for R. (c) A given galvanometer has a resistance of $0.80\ \Omega$ and gives full scale deflection for $1.0\ \mathrm{A}$. Calculate the exact value of the resistance required to convert it into an ammeter capable of measuring up to $10.0\ \mathrm{A}$. [Practice] Very High

Answer Key

Q1: Along the negative z-axis (using Fleming's Left-Hand Rule or $\vec{F} = -e(\hat{i} \times \hat{j}) = -e\hat{k}$). **Q2:** $S = \frac{I_g R_g}{I - I_g} = \frac{1 \times 0.81}{10 - 1} = \frac{0.81}{9} = 0.09\ \Omega$. **Q3:** Magnetic moment $M = \frac{e}{2m} L$. **Q4:** When the wire is strictly perpendicular to the magnetic field ($\theta = 90^\circ$). **Q5:** $mg = BIl \implies I = \frac{mg}{Bl}$. **Q6:** It will move in a straight line. (Magnetic force is zero as $\theta = 0$; electric force accelerates it linearly). **Q7:** $r = \frac{\sqrt{2mVq}}{qB} \implies r \propto \sqrt{V}$. If V is doubled, new radius $r' = \sqrt{2}r$. **Q8:** Independent of the speed (or kinetic energy) of the particle ($T = \frac{2\pi m}{qB}$). **Q9:** Inside ($r < a$): $B \propto r \implies B_1 = B_{surf}/2$. Outside ($r > a$): $B \propto 1/r \implies B_2 = B_{surf}/2$. Ratio is $1:1$. **Q10:** Perimeter L is same. Square $a = L/4 \implies A_s = L^2/16$. Circle $r = L/2\pi \implies A_c = L^2/4\pi$. $M_s/M_c = A_s/A_c = \pi : 4$. **Q11:** $B_1 = \frac{\mu_0 I}{2R}$. $B_2 = \frac{\mu_0 I R^2}{2(R^2+R^2)^{3/2}} = \frac{\mu_0 I}{4\sqrt{2}R}$. Ratio $B_1/B_2 = 2\sqrt{2} : 1$. **Q12:** $B_p = B_q \implies \frac{\mu_0 I_p}{2(2x)} = \frac{\mu_0 (9)}{2(3x)} \implies \frac{I_p}{2} = 3 \implies I_p = 6\ \mathrm{A}$. Direction: Clockwise. **Q13:** Connect a very high resistance in series with the ammeter. **Q14:** (d) material of the turns of the coil. **Q15:** $T = \frac{2\pi m}{qB}$. $T_p \propto \frac{m}{e}$. $T_\alpha \propto \frac{4m}{2e} = \frac{2m}{e}$. Ratio $T_p : T_\alpha = 1:2$. **Q16:** $\vec{F} = q(\vec{v} \times \vec{B})$. **Q17:** 1 Tesla is the magnetic field in which a charge of 1 Coulomb moving with a speed of 1 m/s perpendicularly to the field experiences a force of 1 Newton. **Q18:** (a) Both A and R are true, and R is the correct explanation. ($M = I(\pi r^2) \propto r^2$). **Q19:** (a) Both A and R are true, and R is the correct explanation. **Q20:** (a) Both A and R are true, and R is the correct explanation. (Parallel shunt lowers the equivalent resistance). **Q21:** $B_{net} = \sqrt{B_P^2 + B_Q^2}$. $B \propto I \implies B_{net} \propto \sqrt{3^2 + 4^2} = 5$. Magnitude corresponds to 5A equivalent. $B_{net} = \frac{\mu_0 (5)}{2(0.05)} = 2\pi \times 10^{-5}\ \mathrm{T}$. Direction is at $\tan^{-1}(4/3)$ to the plane of P. **Q22:** $M = NIA = 2000 \times 4.0 \times 1.6 \times 10^{-4} = 1.28\ \mathrm{Am}^2$. **Q23:** $mg = BIl \implies B = \frac{mg}{Il} = \frac{0.2 \times 9.8}{2 \times 1.5} = 0.653\ \mathrm{T}$. **Q24:** Electric force must exactly cancel magnetic force: $qE = qvB \implies v = E/B$, with $\vec{v}$, $\vec{E}$, and $\vec{B}$ being mutually perpendicular. **Q25:** $r = \frac{\sqrt{2mK}}{qB}$. $r_p \propto \frac{\sqrt{1}}{1} = 1$. $r_\alpha \propto \frac{\sqrt{4}}{2} = 1$. Ratio $r_p : r_\alpha = 1:1$. **Q26:** Deflection produced per unit current ($I_s = \theta/I = NAB/k$). SI unit: rad/A or div/A. **Q27:** $S = \frac{I_g R_g}{I - I_g} = \frac{1.0 \times 0.8}{5.0 - 1.0} = \frac{0.8}{4} = 0.2\ \Omega$. **Q28:** $f = \frac{\mu_0 I_1 I_2}{2\pi d} = \frac{4\pi \times 10^{-7} \times 5 \times 5}{2\pi \times 1} = 5 \times 10^{-6}\ \mathrm{N/m}$. Attractive. **Q29:** $\tau = NIAB \sin\theta = 20 \times 12 \times (0.1)^2 \times 0.80 \times \sin 30^\circ = 20 \times 12 \times 0.01 \times 0.80 \times 0.5 = 0.96\ \mathrm{Nm}$. **Q30:** Zero. (Magnetic force is always perpendicular to velocity, so $\vec{F} \cdot d\vec{s} = 0$). **Q31:** Electron. Frequency $\nu = \frac{qB}{2\pi m}$. Since $m_e \ll m_p$, $\nu_e \gg \nu_p$. **Q32:** To ensure the voltmeter draws negligible current from the main circuit, preventing it from altering the potential difference it is trying to measure. **Q33:** $B_{net} = \sqrt{B_P^2 + B_Q^2} = \frac{\mu_0}{2R}\sqrt{1^2 + (\sqrt{3})^2} = \frac{\mu_0}{2R}\sqrt{4} = \frac{\mu_0}{R}$. **Q34:** (i) $r < a : \oint B \cdot dl = \mu_0 I_{encl} \implies B(2\pi r) = \mu_0 I (r^2/a^2) \implies B = \frac{\mu_0 I r}{2\pi a^2}$. (ii) $r > a : B(2\pi r) = \mu_0 I \implies B = \frac{\mu_0 I}{2\pi r}$. **Q35:** (i) KE $= qV$. Ratio $1:1:2$. (ii) $r = \sqrt{2mK}/qB \implies r \propto \sqrt{m/q}$. $r_p \propto \sqrt{1/1} = 1 \implies 5\ \mathrm{cm}$. $r_d \propto \sqrt{2/1} \implies 5\sqrt{2}\ \mathrm{cm}$. $r_\alpha \propto \sqrt{4/2} = \sqrt{2} \implies 5\sqrt{2}\ \mathrm{cm}$. **Q36:** Trajectory is a helix. Circular motion due to $v \sin 30^\circ$, linear motion due to $v \cos 30^\circ$. Pitch $P = v_{||} \times T = (v \cos 30^\circ)\frac{2\pi m}{qB}$. **Q37:** Wires attract due to Lorentz force $F = I_2 L B_1$. $F/L = \frac{\mu_0 I_1 I_2}{2\pi d}$. 1 Ampere is the steady current that, flowing in two infinitely long parallel wires 1m apart in vacuum, produces a force of $2 \times 10^{-7}\ \mathrm{N/m}$. **Q38:** Derive $dB = \frac{\mu_0 I dl \sin 90^\circ}{4\pi(R^2+x^2)}$. Resolve components; $\cos\theta$ components cancel, $\sin\theta$ components add up. Integrate to get $B = \frac{\mu_0 I R^2}{2(R^2+x^2)^{3/2}}$. **Q39:** Draw rectangular Amperian loop $abcd$ inside and outside. $\oint B \cdot dl = \int_a^b Bdl = Bl$. Current enclosed $= nlI$. $Bl = \mu_0 nlI \implies B = \mu_0 nI$. **Q40:** (i) At midpoint, fields are equal and opposite, $B_{net} = 0$. (ii) Graph approaches $+\infty$ near A, crosses x-axis at $d/2$, and approaches $-\infty$ near B. **Q41:** Force on near arm is attractive and stronger (closer distance). Force on far arm is repulsive and weaker. Side arms cancel. Net force is attractive. **Q42:** (1) It has finite resistance, which would lower the circuit current. (2) It's very sensitive and burns out at normal currents. Converted by adding a low resistance shunt in parallel. **Q43:** (A) A current-carrying coil in a magnetic field experiences a torque. (B) $\tau = NIAB\sin\theta$. (C) Radial B makes $\theta = 90^\circ$ always, so $\tau = NIAB$ (linear scale). Achieved by cylindrical pole pieces and a soft iron core. **Q44:** (A) $\vec{\tau} = \vec{m} \times \vec{B}$. (B) Stable: $\vec{m}$ parallel to $\vec{B}$ ($\theta = 0^\circ$). Unstable: $\vec{m}$ antiparallel to $\vec{B}$ ($\theta = 180^\circ$). (C) $\tau_{max} = NIAB = 20 \times 5 \times 0.01 \times 0.2 = 0.2\ \mathrm{Nm}$. **Q45:** (A) $\vec{F} = q(\vec{v} \times \vec{B})$. (B) (i) $\vec{v} \perp \vec{B}$. (ii) $\vec{v}$ at an oblique angle to $\vec{B}$. (C) $\vec{F} \perp \vec{v}$, so work done $W = \vec{F} \cdot d\vec{s} = 0 \implies \Delta KE = 0$. **Q46:** (A) Let x be distance from I. $\frac{\mu_0 I}{2\pi x} = \frac{\mu_0 (2I)}{2\pi(d-x)} \implies \frac{1}{x} = \frac{2}{d-x} \implies d - x = 2x \implies x = d/3$. (B) No, the net force relies entirely on the external fields cancelling out at that spot, making $I_3 \times B_{net} = 0$ regardless of I_3. **Q47:** (A) Similarities: Both are long-range, inverse square laws, obey superposition. Differences: Electrostatic is produced by a scalar (charge), magnetic by a vector (Idl). Electrostatic acts along position vector, magnetic is perpendicular to plane of Idl and r. (B) Integrate $dB = \frac{\mu_0 I dl}{4\pi R^2}$ over circle length $2\pi R \implies B = \frac{\mu_0 I}{2R}$. **Q48:** (A) $I_s = \theta/I$. $V_s = \theta/V$. (B) $V_s = \frac{\theta}{IR} = \frac{I_s}{R}$. (C) No. Doubling turns doubles I_s, but it also doubles the wire length and hence the resistance R. Since $V_s = I_s/R$, it remains unchanged. **Q49:** (A) $KE = qV \implies KE_\alpha/KE_p = 2qV/qV = 2:1$. (B) $v = \sqrt{2K/m} \implies v_\alpha/v_p = \sqrt{2/4} = 1:\sqrt{2}$. (C) $r = \sqrt{2mK}/qB \implies r_\alpha/r_p = \frac{\sqrt{4\times2}}{2}/\frac{\sqrt{1\times1}}{1} = \frac{\sqrt{8}}{2} : 1 = \sqrt{2}:1$. (D) $T = 2\pi m/qB \implies T_\alpha/T_p = (4/2)/(1/1) = 2:1$. **Q50:** (a) $I_g R_g = (I - I_g)S \implies S = \frac{I_g R_g}{I - I_g}$. (b) $V = I_g(R_g + R) \implies R = \frac{V}{I_g} - R_g$. (c) $S = \frac{1.0\times0.80}{10.0-1.0} = \frac{0.80}{9} = 0.088\ \Omega$.
